GIT AND GITHUB PART-2

1.Install Git

Step 1: Download Git

1. Open your web browser.
2. Go to the official Git website: https://git-scm.com/downloads
3. Click **Download for Windows**.

Step 2: Run the Installer

1. Double-click the downloaded .exe file.
2. Click **Yes** if prompted

Step 4: Open Git Bash

After installation:

- Press **Windows**
- Search for **Git Bash**
- Open **Git Bash**

Verify git installation: $ git --version

2.Create a repo in github with README.md and .ignore file.

Objective: To create a new GitHub repository initialized with a README.md file and a .gitignore file for managing project documentation and ignoring unnecessary files.

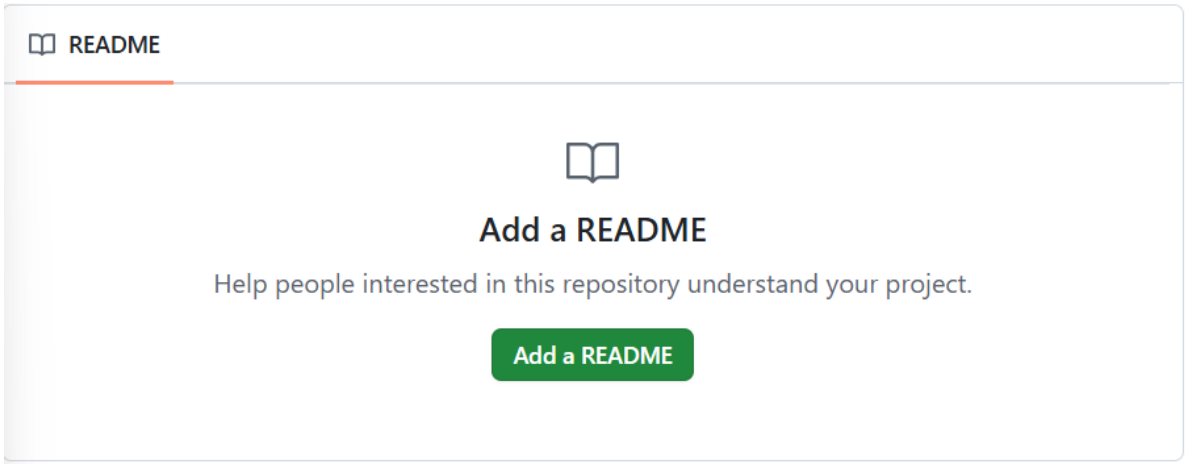
Purpose: The purpose of creating a GitHub repository with a README.md and .gitignore file is to organize the project, provide project information, and prevent unwanted files from being tracked by Git.

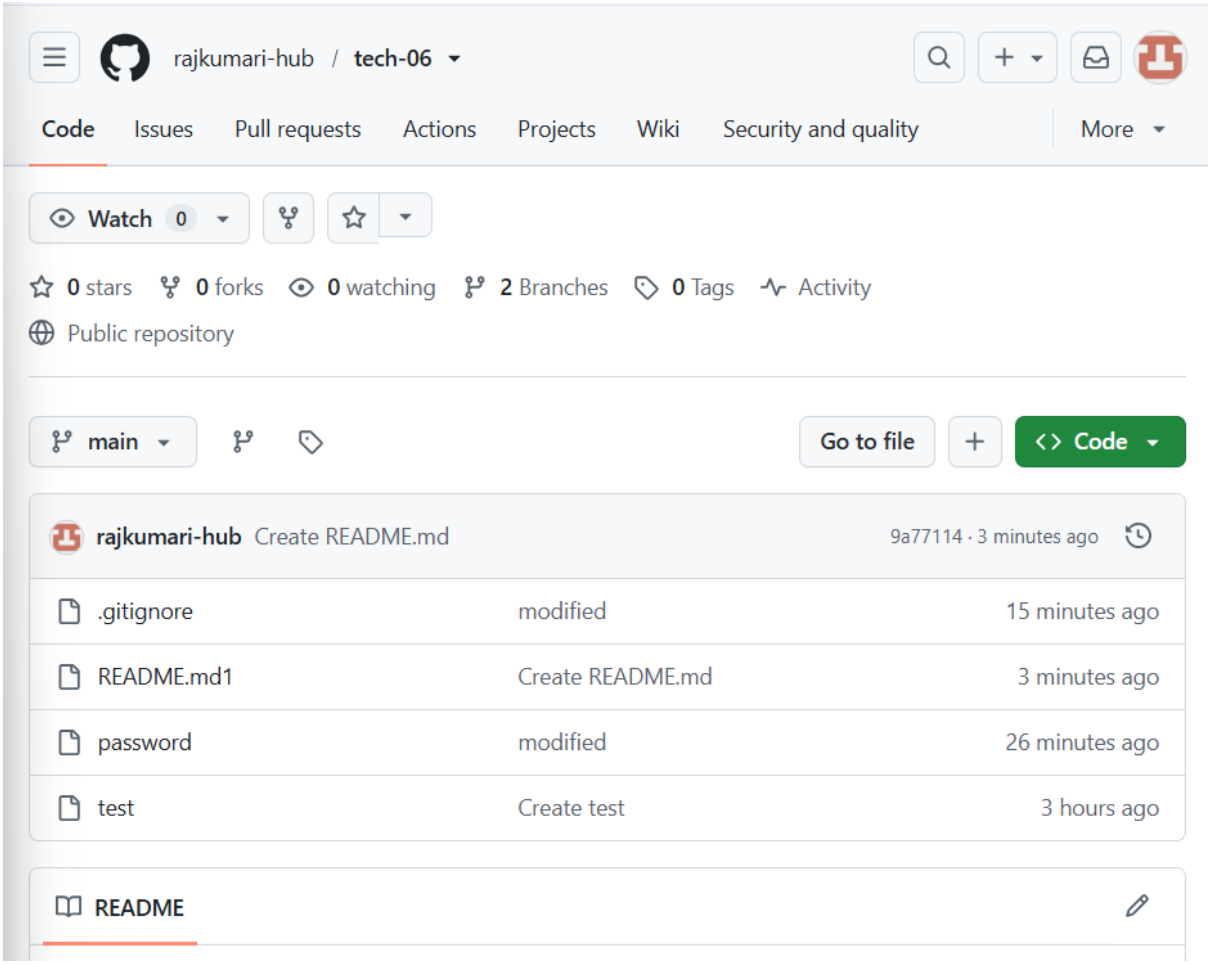
Step 1: Sign in to GitHub

1. **Open your web browser.**
2. **Go to https://github.com**
3. **Sign in with your GitHub account.**

Step 2: Create a New Repository

1. **Click the + icon in the top-right corner.**
2. **Select New repository.**





3.Clone the created repo to local.

Objective: To clone an existing GitHub repository from the remote server to the local machine using Git.

Purpose: The purpose of cloning a repository is to create a complete local copy of the GitHub repository, including all

files, folders, branches, and commit history, so you can work on the project locally.

Step 1: Navigate to the Destination Folder

Move to the folder where you want to store the repository

Step 2: Clone the Repository

Run the following command:

$ git clone git@github.com:rajkumari-hub/tech-06.git

If we want to clone with separate branch, run the following command

$ git clone -b main git@github.com:rajkumari-hub/tech-06.git

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop
$ git clone -b main git@github.com:rajkumari-hub/tech-06.git
Cloning into 'tech-06'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.

brajk@RajKumari MINGW64 ~/OneDrive/Desktop
$ cd tech-06

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ ls
test

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ git branch --list
* main

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$
```

4.Create two files in local repo

Using touch command we can create two files

$ touch developer.txt engineer.txt

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ touch developer.txt engineer.txt

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ ls
README.md1  developer.txt  devops  engineer.txt  password  test

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$
```

5.Commit two files and push to central Repository.

Step 1: After creating two files, Check the status for untracking files

$ git status

Step 2: Add the files to tracking files using command

$ git add .

Step 3: Commit the files

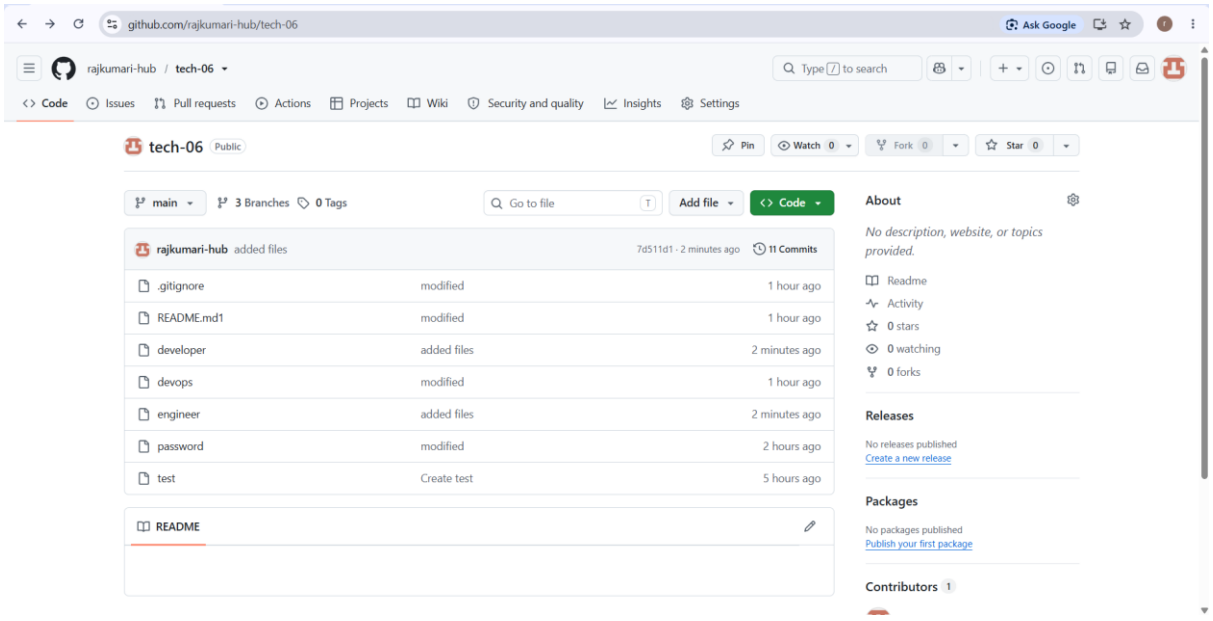
$ git commit -m “Added files”

Step 4: push the files into github repository

$ git push -u origin main

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ ls
README.md1  developer  devops  engineer  password  test

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        developer
        engineer

nothing added to commit but untracked files present (use "git add" to track)

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ git add .

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ git commit -m "added files"
[main 7d511d1] added files
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 developer
 create mode 100644 engineer

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$
```

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 271 bytes | 271.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:rajkumari-hub/tech-06.git
   cc41fa6..7d511d1  main -> main
branch 'main' set up to track 'origin/main'.

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$
```



6. Create a branch in local and create a sample file and push to central.

Step1: create a new branch

$ git branch feature

$ git branch –list

Step2: Switch current branch new branch

$ git checkout feature

Expected o/p: Switched to branch ‘feature’

Step3: Create a sample file

$ touch sample

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ git branch feature

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ git branch --list
  feature
* main

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ git checkout feature
Switched to branch 'feature'

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$ ls
README.md1  developer  devops  engineer  password  test

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$ touch sample

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$
```

Step4: Check the Untracking files

$ git status

$ git add .

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$ ls
README.md1  developer  devops  engineer  password  sample  test

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$ git status
On branch feature
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        sample

nothing added to commit but untracked files present (use "git add" to track)

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$ git add .
```

Step5: Commit the file

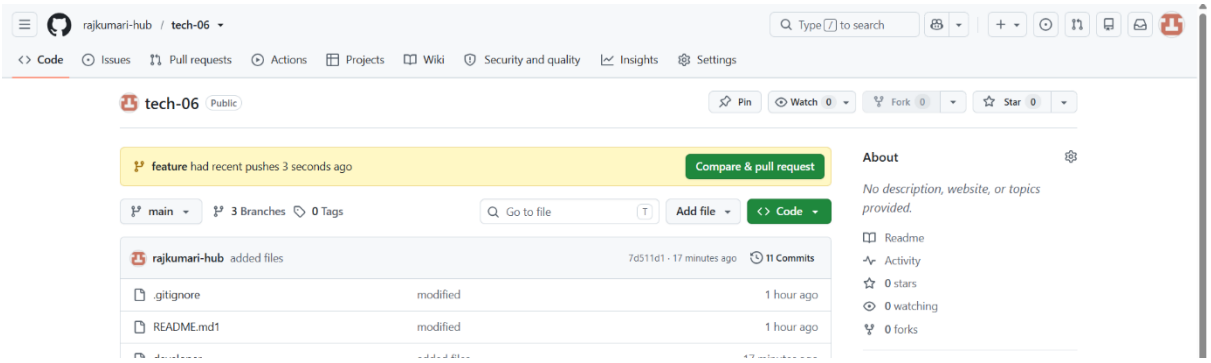
$ git commit -m “add sample file”

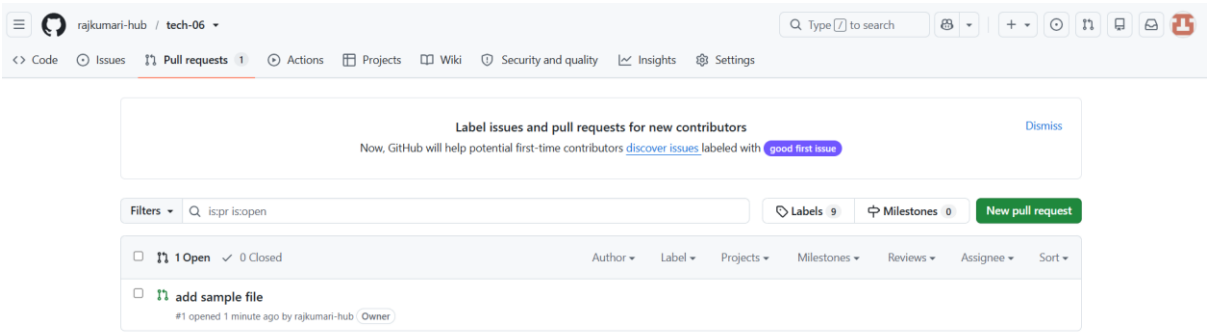
```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$ git status
On branch feature
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   sample


brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$ git commit -m "add sample file"
[feature 38ae343] add sample file
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 sample
```

Step6: push into github repo

$ git push -u origin feature

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$ git push -u origin feature
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 235 bytes | 235.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature' on GitHub by visiting:
remote:      https://github.com/rajkumari-hub/tech-06/pull/new/feature
remote:
To github.com:rajkumari-hub/tech-06.git
 * [new branch]      feature -> feature
branch 'feature' set up to track 'origin/feature'.

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
```



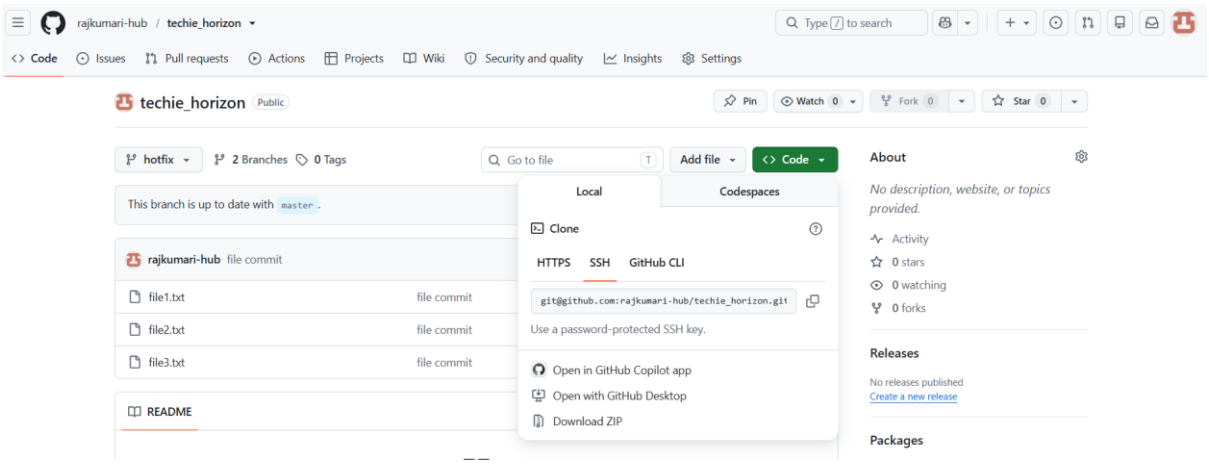


7. Create a branch in github and clone that to local.

Step 1: Create a New Branch

1. Click the **Branch** drop-down in github(it usually shows main).
2. Type the new branch name.

   Ex. hotfix



Step2: Clone into gitbash

$ git clone git@github.com:rajkumari-hub/techie_horizon.git

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop
$ git clone git@github.com:rajkumari-hub/techie_horizon.git
Cloning into 'techie_horizon'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

brajk@RajKumari MINGW64 ~/OneDrive/Desktop
$ ls
 18kq1a0102/                  'Microsoft Edge.lnk'*       hello.py                   techie_horizon/
 DevOps/                       UbuntuShare/               java_tutorial.pdf
'Docker Desktop.lnk'*         'Visual Studio Code.lnk'*  'rajkumari - Chrome.lnk'*
 JavaTheCompleteReference.pdf  desktop.ini                tech-06/

brajk@RajKumari MINGW64 ~/OneDrive/Desktop
$ cd techie_horizon

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/techie_horizon (master)
$ git branch
* master

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/techie_horizon (master)
$ git fetch --all

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/techie_horizon (master)
$ git branch -a
* master
  remotes/origin/HEAD -> origin/master
  remotes/origin/hotfix
  remotes/origin/master

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/techie_horizon (master)
$ git checkout hotfix
Switched to a new branch 'hotfix'
branch 'hotfix' set up to track 'origin/hotfix'.

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/techie_horizon (hotfix)
$ git branch
* hotfix
  master

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/techie_horizon (hotfix)
$ ls
file1.txt  file2.txt  file3.txt

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/techie_horizon (hotfix)
$
```

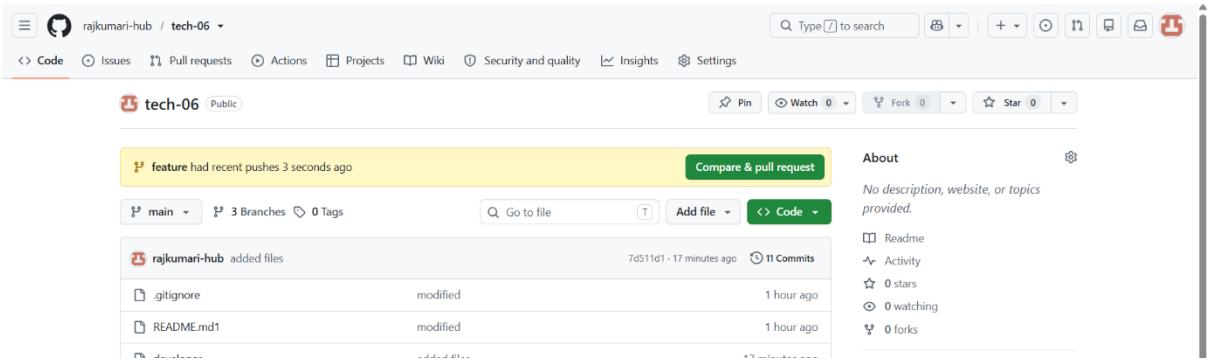
8. Merge the created branch with master in git local.

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/hiring-app (hotfix)
$ git merge release
Already up to date.

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/hiring-app (hotfix)
$ ls
a  b  c  d  docker  e  f  hello  release  z

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/hiring-app (hotfix)
$
```

```
    added z

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/hiring-app (hotfix)
$ git checkout master
Switched to branch 'master'

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/hiring-app (master)
$ ls
docker  hello  master

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/hiring-app (master)
$ git cherry-pick fedb05e4
[master e046ffb] added in hotfix
 Date: Wed Jul 8 12:28:26 2026 +0530
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 hotfix

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/hiring-app (master)
$ ls
docker  hello  hotfix  master

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/hiring-app (master)
$
```

9. Merge the created branch with main in github by sending a pull request.

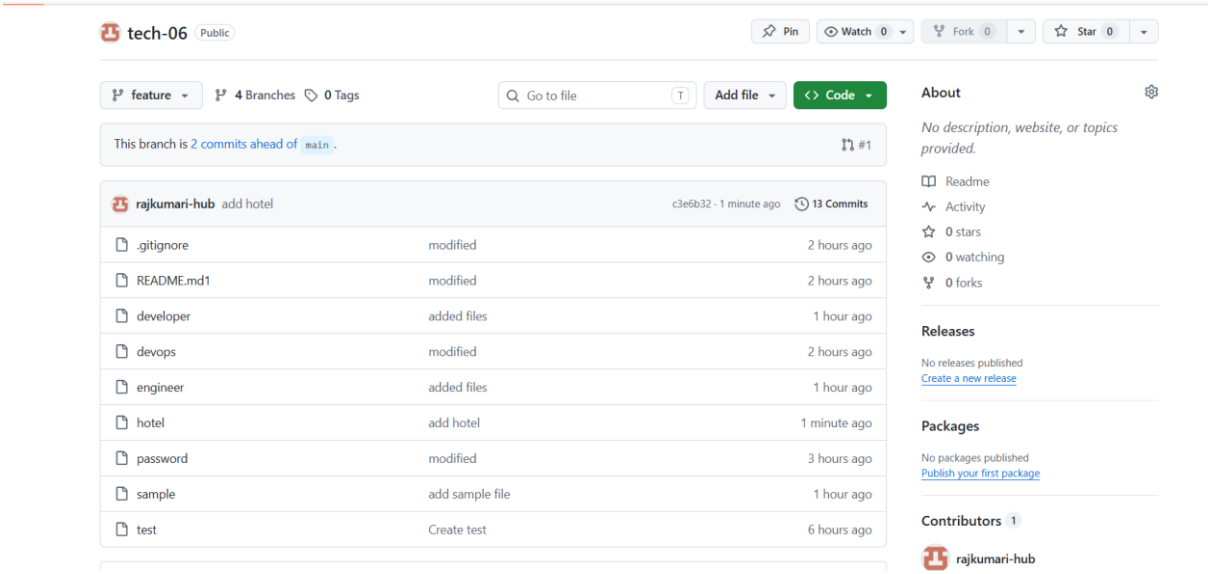
```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$ git branch
* feature
  main

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$ git status
On branch feature
Your branch is up to date with 'origin/feature'.

nothing to commit, working tree clean

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$ git push -u origin feature
Everything up-to-date
branch 'feature' set up to track 'origin/feature'.

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ git push -u origin feature
Everything up-to-date
branch 'feature' set up to track 'origin/feature'.
```



10. Create a file in local and send that to branch in github.

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$ git status
On branch feature
Your branch is up to date with 'origin/feature'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        hotel

nothing added to commit but untracked files present (use "git add" to track)

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$
```

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$ git add .

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$ git commit -m "add hotel"
[feature c3e6b32] add hotel
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 hotel

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$ git status
On branch feature
Your branch is ahead of 'origin/feature' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$ git push -u origin feature
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 230 bytes | 230.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:rajkumari-hub/tech-06.git
   38ae343..c3e6b32  feature -> feature
branch 'feature' set up to track 'origin/feature'.

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (feature)
$
```

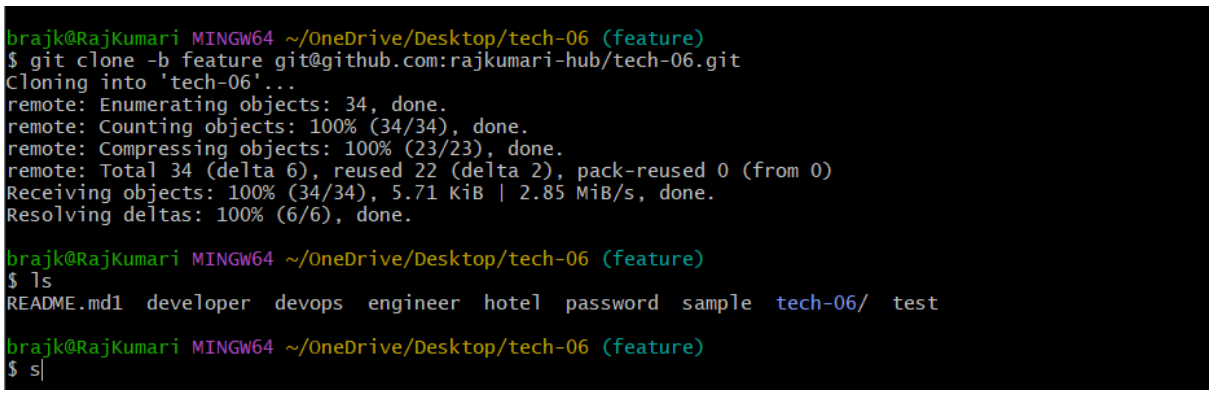


11. Clone only a branch from github to local.

Objective: To clone a specific branch (feature) from a remote GitHub repository to the local machine using the git clone -b command, so that work can begin directly on that branch.

Purpose: The purpose of cloning a specific branch is to download only the desired branch from the remote repository and set it as the active branch in the local repository. This allows developers to work on a particular feature or task without switching branches after cloning.

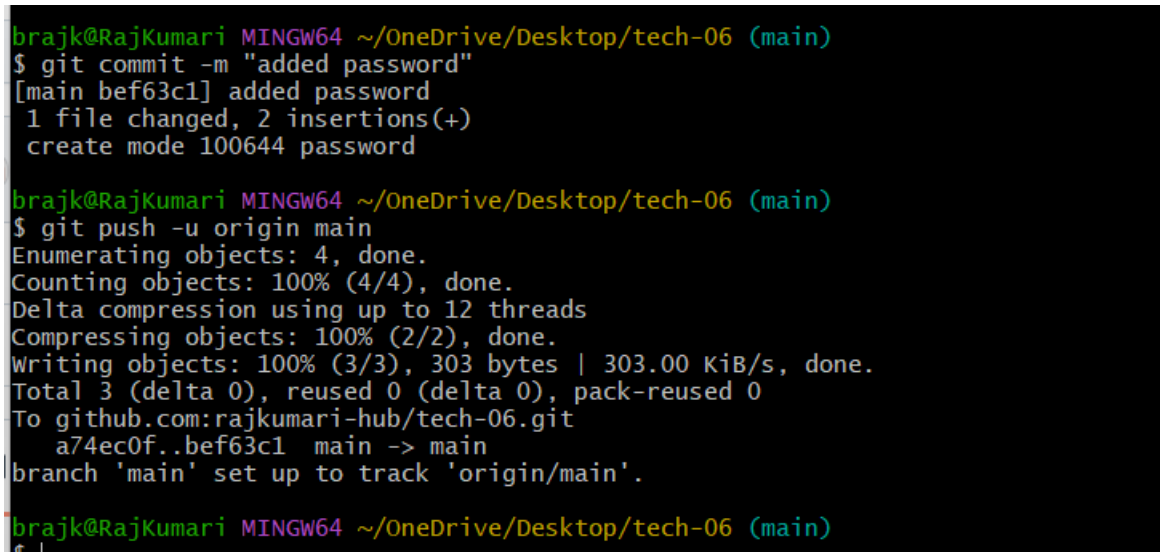
$ git clone -b feature git@github.com:rajkumari-hub/tech-06.git

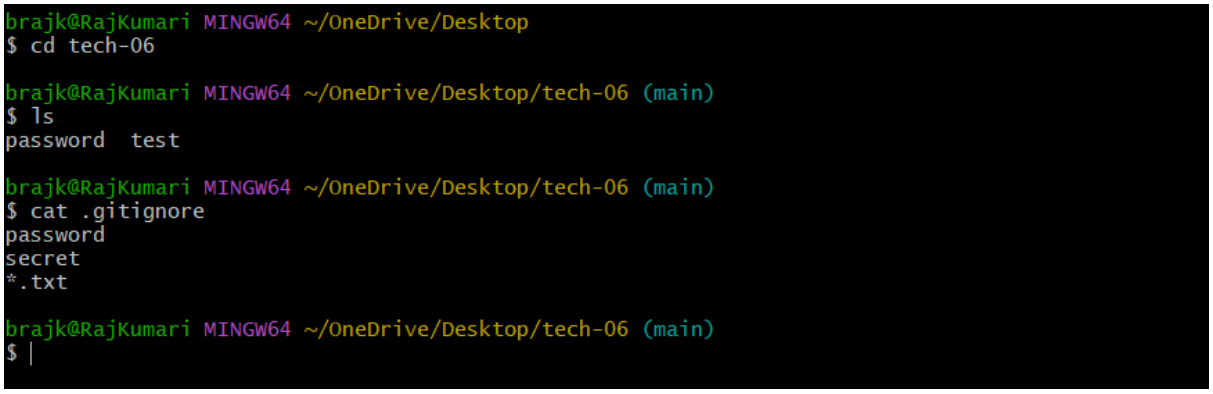


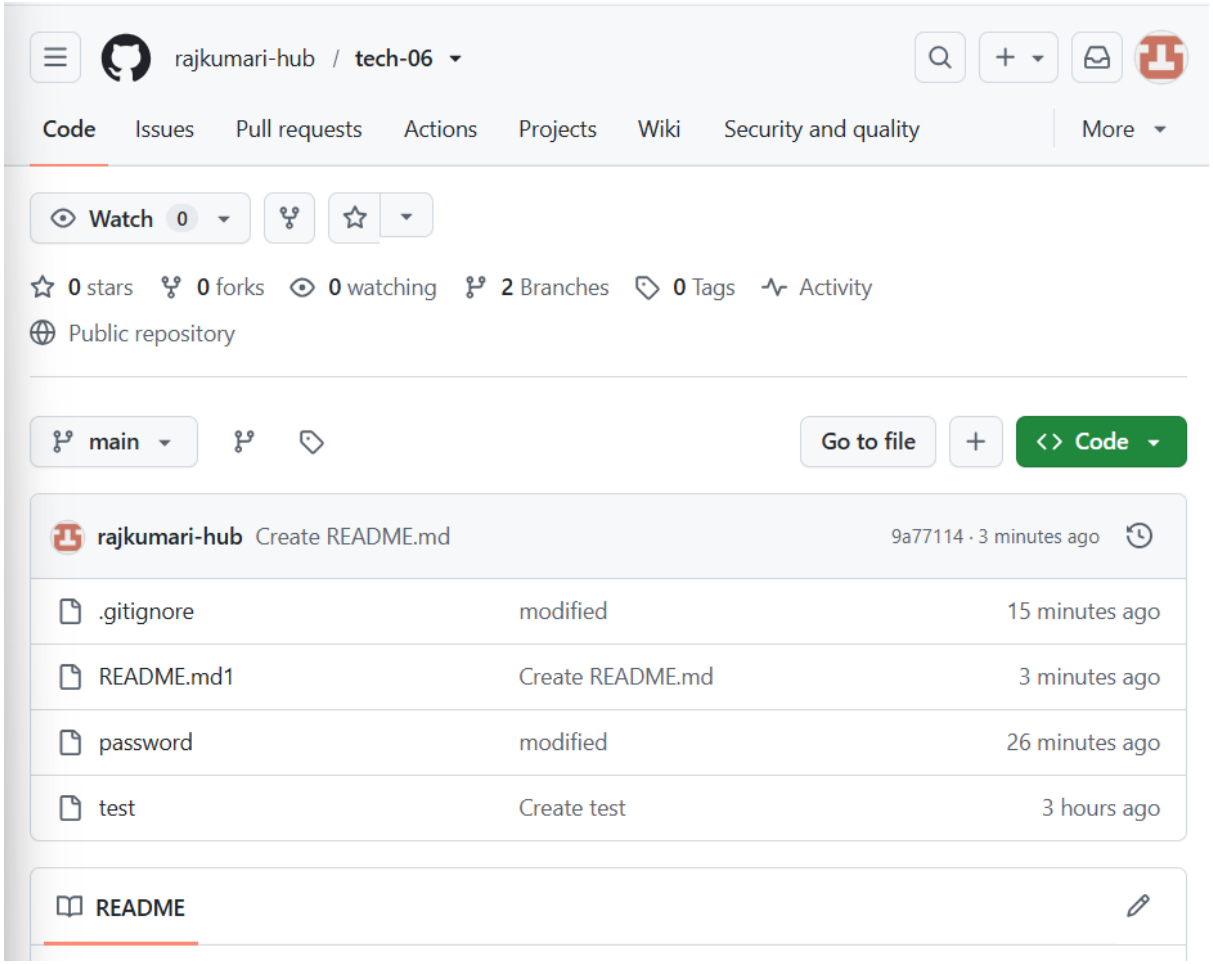
12. Create a file with all passwords and make that untrackable with git.

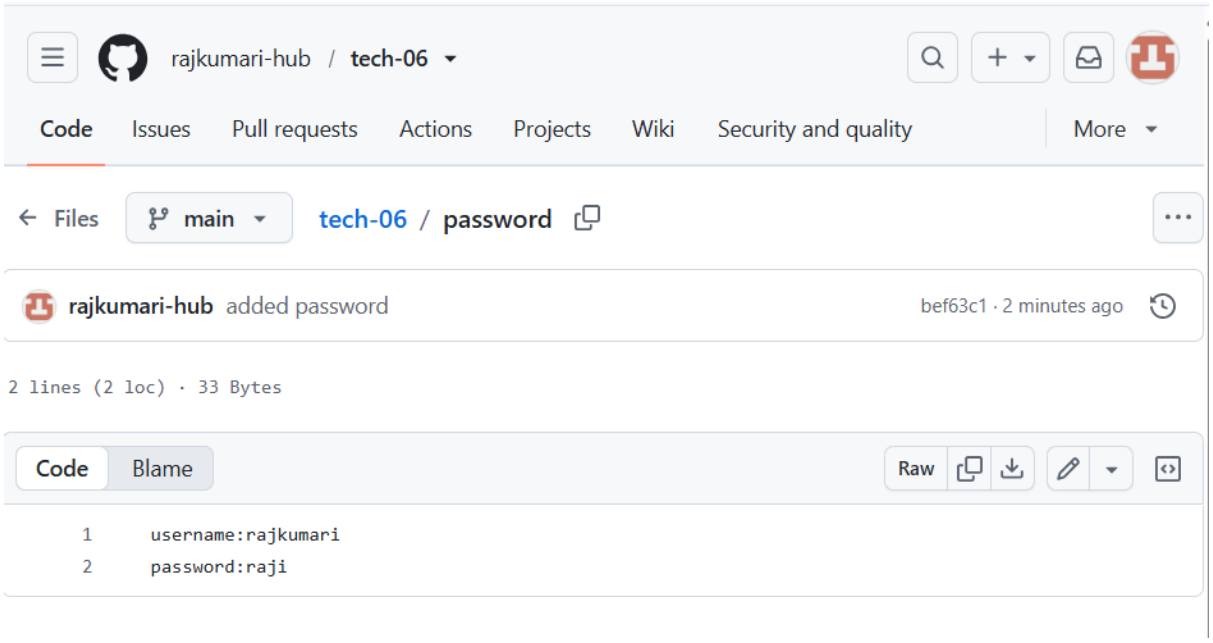
Objective: To create a file containing passwords and configure Git to ignore the file so that it is not tracked or pushed to the GitHub repository.

Purpose: The purpose of this task is to protect sensitive information (such as passwords, API keys, or secrets) by preventing Git from tracking or uploading the file to the remote repository.






rajkumari-hub / tech-06
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
More
Watch 0
0 stars
0 forks
0 watching
2 Branches
0 Tags
Activity
Public repository
main
Go to file
Code
rajkumari-hub Create README.md
9a77114 · 3 minutes ago
.gitignore modified 15 minutes ago
README.md1 Create README.md 3 minutes ago
password modified 26 minutes ago
test Create test 3 hours ago
README


rajkumari-hub / tech-06
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
More
Files
main
tech-06 / password
rajkumari-hub added password
bef63c1 · 2 minutes ago
2 lines (2 loc) · 33 Bytes
Code
Blame
Raw
1 username:rajkumari
2 password:raji

13. Make a commit and make that commit reset without savings changes

Objective: To create a commit and then remove the commit along with all its changes, restoring the repository to the state before the commit.

Purpose: The purpose of git reset --hard is to permanently discard the most recent commit and all the changes associated with it. This is useful when a commit was made by mistake and you no longer need the changes.

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06/tech-06 (feature)
$ ls
README.md1  developer  devops  engineer  hotel  password  sample  test

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06/tech-06 (feature)
$ touch restfile

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06/tech-06 (feature)
$ ls
README.md1  developer  devops  engineer  hotel  password  restfile  sample  test

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06/tech-06 (feature)
$ git status
On branch feature
Your branch is up to date with 'origin/feature'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        restfile

nothing added to commit but untracked files present (use "git add" to track)

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06/tech-06 (feature)
$ git add .

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06/tech-06 (feature)
$ git commit -m "add reset file"
[feature bed51da] add reset file
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 restfile

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06/tech-06 (feature)
$ git log --oneline
bed51da (HEAD -> feature) add reset file
c3e6b32 (origin/feature) add hotel
38ae343 add sample file
7d511d1 (origin/main, origin/HEAD) added files
cc41fa6 Merge branch 'main' of github.com:rajkumari-hub/tech-06
7bdc522 modified
0a45bd9 (origin/test) Create devops
9a77114 Create README.md
2e71ac7 modified
737c10a modified
a7a4411 modified
88489db added .gitignore
bef63c1 added password
a74ec0f Create test

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06/tech-06 (feature)
$
```

Here, Reset the Last Commit Without Saving Changes by using command

$ git reset --hard HEAD~1

For checking

$ git log –oneline

Here, there no untracking file

$ git status

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06/tech-06 (feature)
$ git reset --hard HEAD~1
HEAD is now at c3e6b32 add hotel

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06/tech-06 (feature)
$ git log --oneline
c3e6b32 (HEAD -> feature, origin/feature) add hotel
38ae343 add sample file
7d511d1 (origin/main, origin/HEAD) added files
cc41fa6 Merge branch 'main' of github.com:rajkumari-hub/tech-06
7bdc522 modified
0a45bd9 (origin/test) Create devops
9a77114 Create README.md
2e71ac7 modified
737c10a modified
a7a4411 modified
88489db added .gitignore
bef63c1 added password
a74ec0f Create test

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06/tech-06 (feature)
$ git status
On branch feature
Your branch is up to date with 'origin/feature'.

nothing to commit, working tree clean

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06/tech-06 (feature)
$
```

14. Revert a commited commit to the older version

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06/tech-06 (feature)
$ git revert 38ae343
[feature dc8db46] Revert "add sample file"
 1 file changed, 0 insertions(+), 0 deletions(-)
 delete mode 100644 sample

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06/tech-06 (feature)
$ git log --oneline
dc8db46 (HEAD -> feature) Revert "add sample file"
c3e6b32 (origin/feature) add hotel
38ae343 add sample file
7d511d1 (origin/main, origin/HEAD) added files
cc41fa6 Merge branch 'main' of github.com:rajkumari-hub/tech-06
7bdc522 modified
0a45bd9 (origin/test) Create devops
9a77114 Create README.md
2e71ac7 modified
737c10a modified
a7a4411 modified
88489db added .gitignore
bef63c1 added password
a74ec0f Create test

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06/tech-06 (feature)
$
```

```
Revert "add sample file"

This reverts commit 38ae343ffe5f19ffa64c332c1eee23c32e23e132.

# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# On branch feature
# Your branch is up to date with 'origin/feature'.
#
# Changes to be committed:
#       deleted:    sample
#
~
~
```

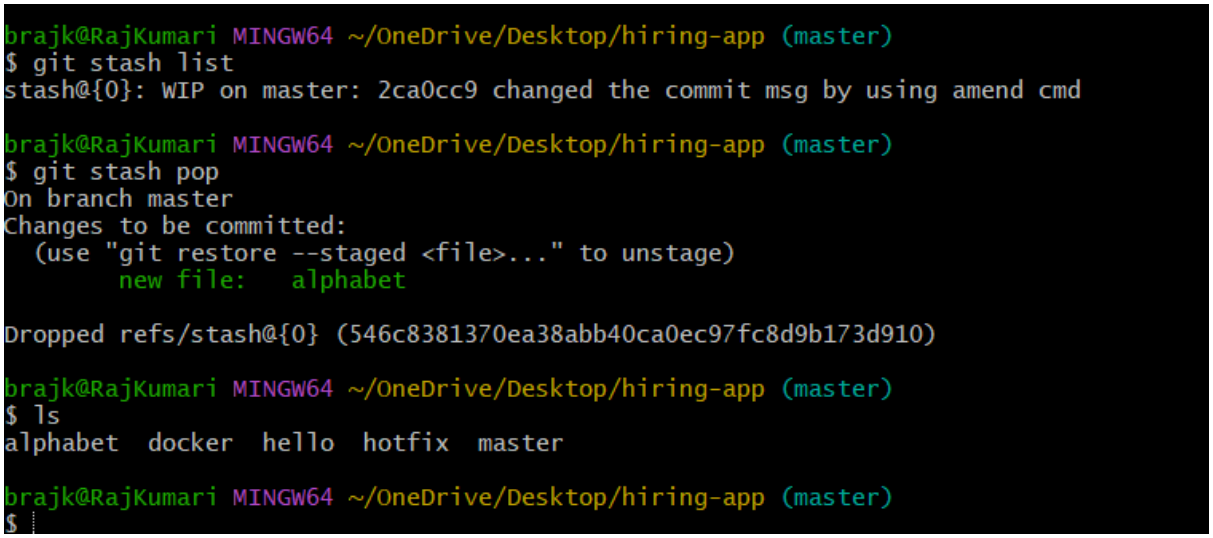
```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06/tech-06 (feature)
$ git status
On branch feature
Your branch is ahead of 'origin/feature' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06/tech-06 (feature)
$
```

15. Push a file to stash without savings the changes and work on another file.

Objective: To temporarily save uncommitted changes using Git Stash, switch to another task, and continue working on a different file without committing the current changes.

Purpose: The purpose of Git Stash is to temporarily store uncommitted changes in a stack, allowing you to work on another file or branch without losing your current work. You can restore the stashed changes later whenever needed.

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/hiring-app (master)
$ ls
alphabet  docker  hello  hotfix  master

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/hiring-app (master)
$ git stash
Saved working directory and index state WIP on master: 2ca0cc9 changed the commit ms
g by using amend cmd

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/hiring-app (master)
$ ls
docker  hello  hotfix  master

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/hiring-app (master)
$
```



16. Undo the stash file and start working on that again.

Objective: To restore the previously stashed changes back to the working directory and continue working on the same file.

Purpose: The purpose of undoing a stash is to retrieve temporarily saved changes and resume work without losing any modifications. Git provides git stash apply and git stash pop to restore stashed changes.

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ ls
README.md1  developer  devops  engineer  password  resetfile  tech-06/  test

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ echo "new content" >> developer

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   developer

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        resetfile
        tech-06/

no changes added to commit (use "git add" and/or "git commit -a")

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ git stash
warning: in the working copy of 'developer', LF will be replaced by CRLF the next time Git touches it
Saved working directory and index state WIP on main: 7d511d1 added files
```

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ git stash list
stash@{0}: WIP on main: 7d511d1 added files

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ git stash pop
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   developer

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        resetfile
        tech-06/

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (ea647b5a582602c5837ac72ba01c4ce2b7aa0056)

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.
```

17. Generate a ssh-keygen and configure into github.

Objective: To generate an SSH key pair on the local machine and configure the public key in GitHub to enable secure authentication without entering a username and password for every Git operation.

Purpose: The purpose of using SSH keys is to establish a secure connection between the local machine and GitHub. Once configured, you can clone, push, and pull repositories using SSH authentication.

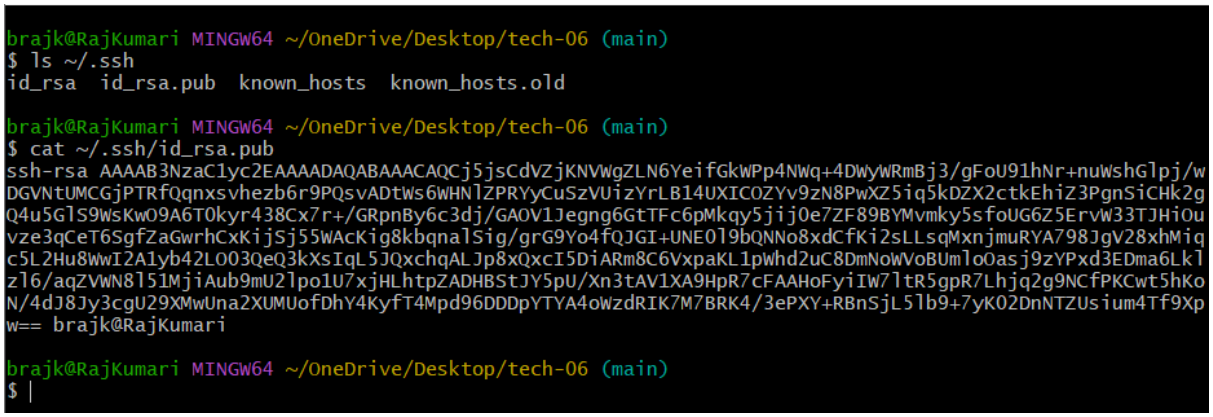
Step 1: Generate a New SSH Key

$ ssh-keygen

Check, if already exist

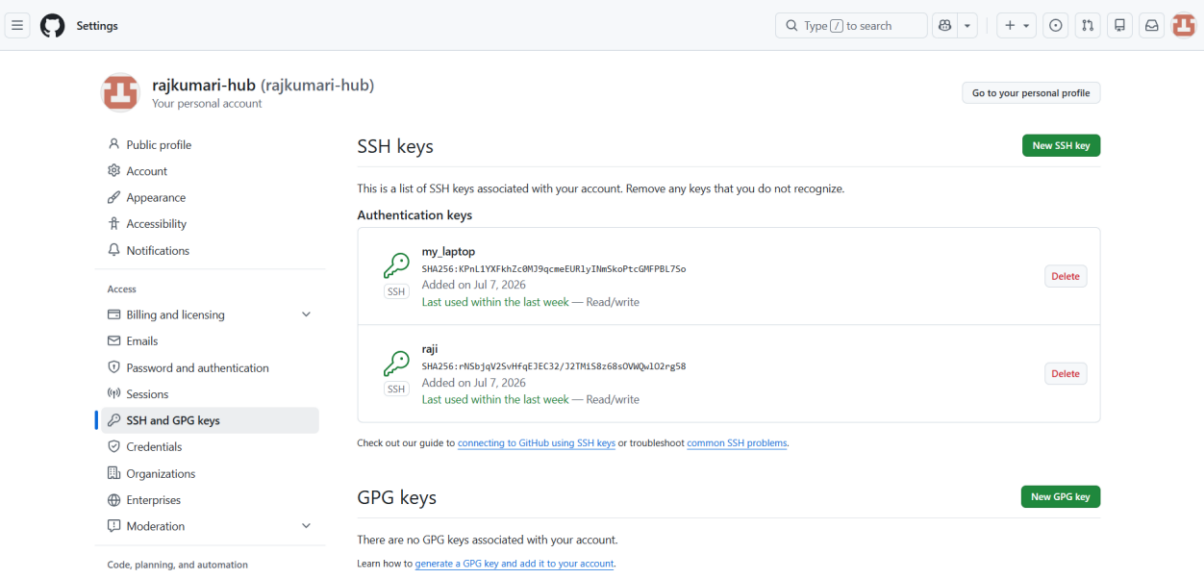
$ ls ~/.ssh

$ cat ~/.ssh/id_rsa.pub



Step2: **Add the SSH Key to GitHub**

1. Log in to **GitHub**.
2. Click your **Profile** icon.
3. Select **Settings**.
4. Click **SSH and GPG keys**.
5. Click **New SSH key**.
6. Enter a title, for example



$ git remote add origin git@github.com:rajkumari-hub/tech-06.git

$ git remote -v

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ git remote -v
origin  git@github.com:rajkumari-hub/tech-06.git (fetch)
origin  git@github.com:rajkumari-hub/tech-06.git (push)

brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$
```

18. Configure webhooks to github.

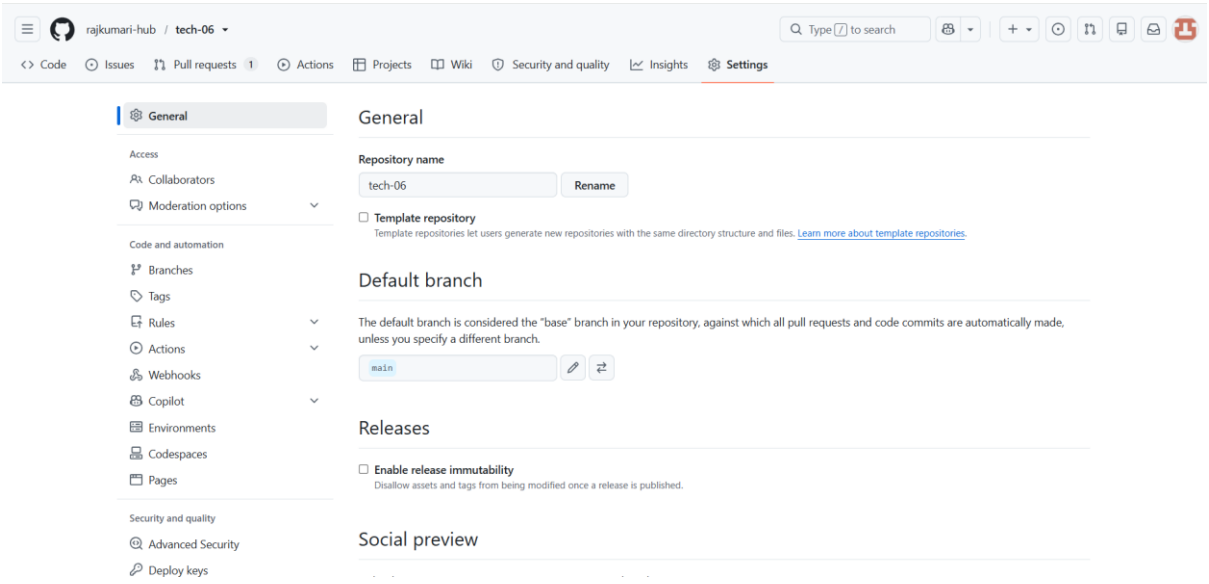
Objective: To configure a GitHub Webhook so that GitHub automatically sends event notifications (such as push, pull request, or release events) to an external application whenever specific actions occur in a repository.

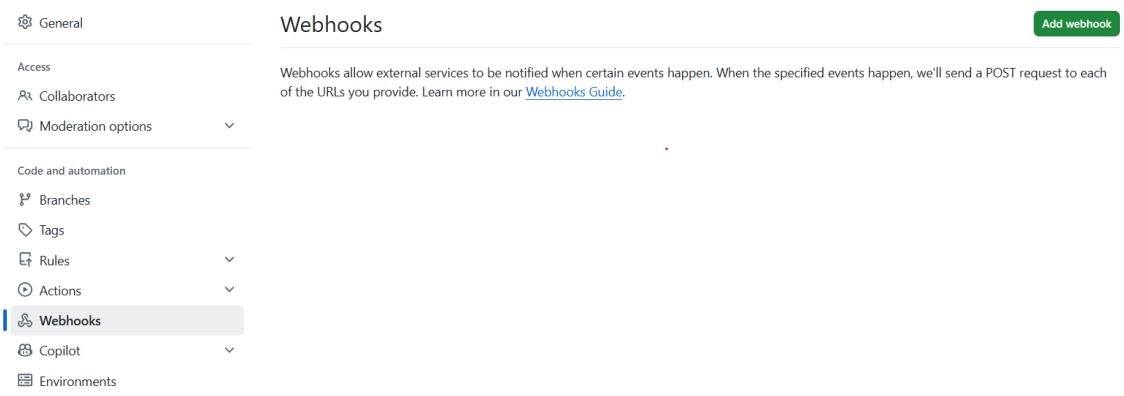
Purpose: GitHub Webhooks automate workflows by notifying external services whenever an event occurs in a GitHub repository. They are commonly used for Continuous Integration (CI), Continuous Deployment (CD), Jenkins, Slack notifications, deployment pipelines, and custom applications.

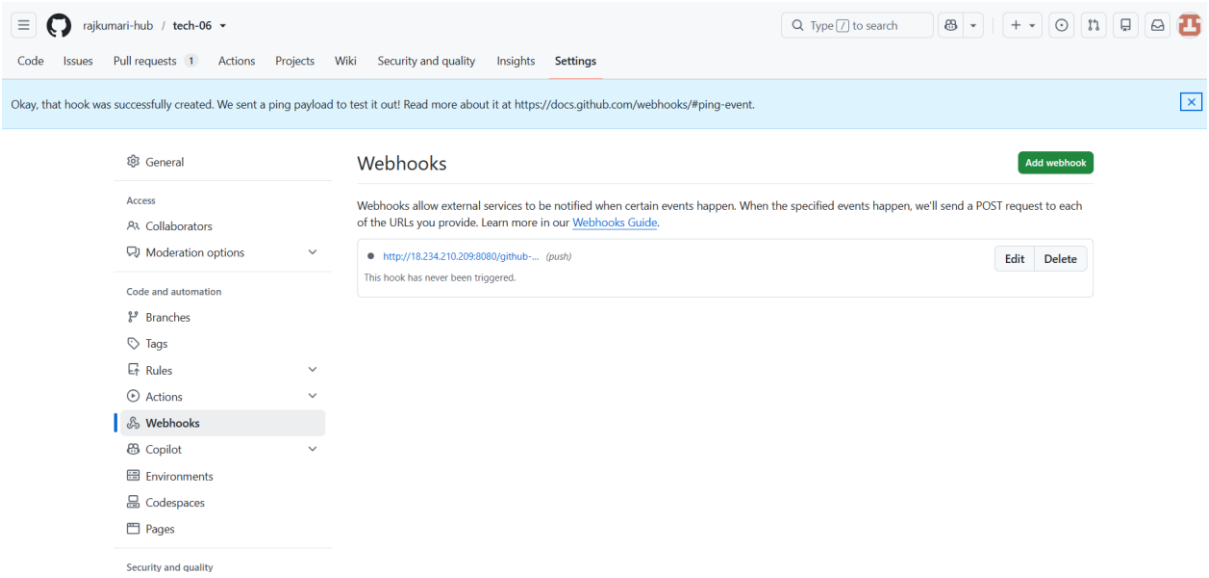
Steps

1. Sign in to your GitHub account.
2. Open the repository where you want to configure the webhook.
3. Click **Settings**.
4. In the left sidebar, click **Webhooks**.
5. Click **Add webhook**.
6. In the **Payload URL** field, enter the URL of your application (the endpoint that will receive webhook events).
7. Select **Content type** as **application/json**.
8. (Optional) Enter a **Secret** to secure the webhook.
9. Choose the events that should trigger the webhook:
    - **Just the push event** (recommended), or
    - **Send me everything**, or
    - **Let me select individual events**.
10. Ensure the **Active** checkbox is selected.
11. Click **Add webhook** to save the configuration.
12. Push a commit or perform the selected event to test the webhook.
13. Verify the webhook delivery by going to **Settings → Webhooks → Select the webhook → Recent Deliveries**.

A **200 OK** response indicates the webhook is working successfully.







19. Basic understanding of .git file

Objective: To understand the purpose of the .git folder and how Git uses it to manage version control for a repository.

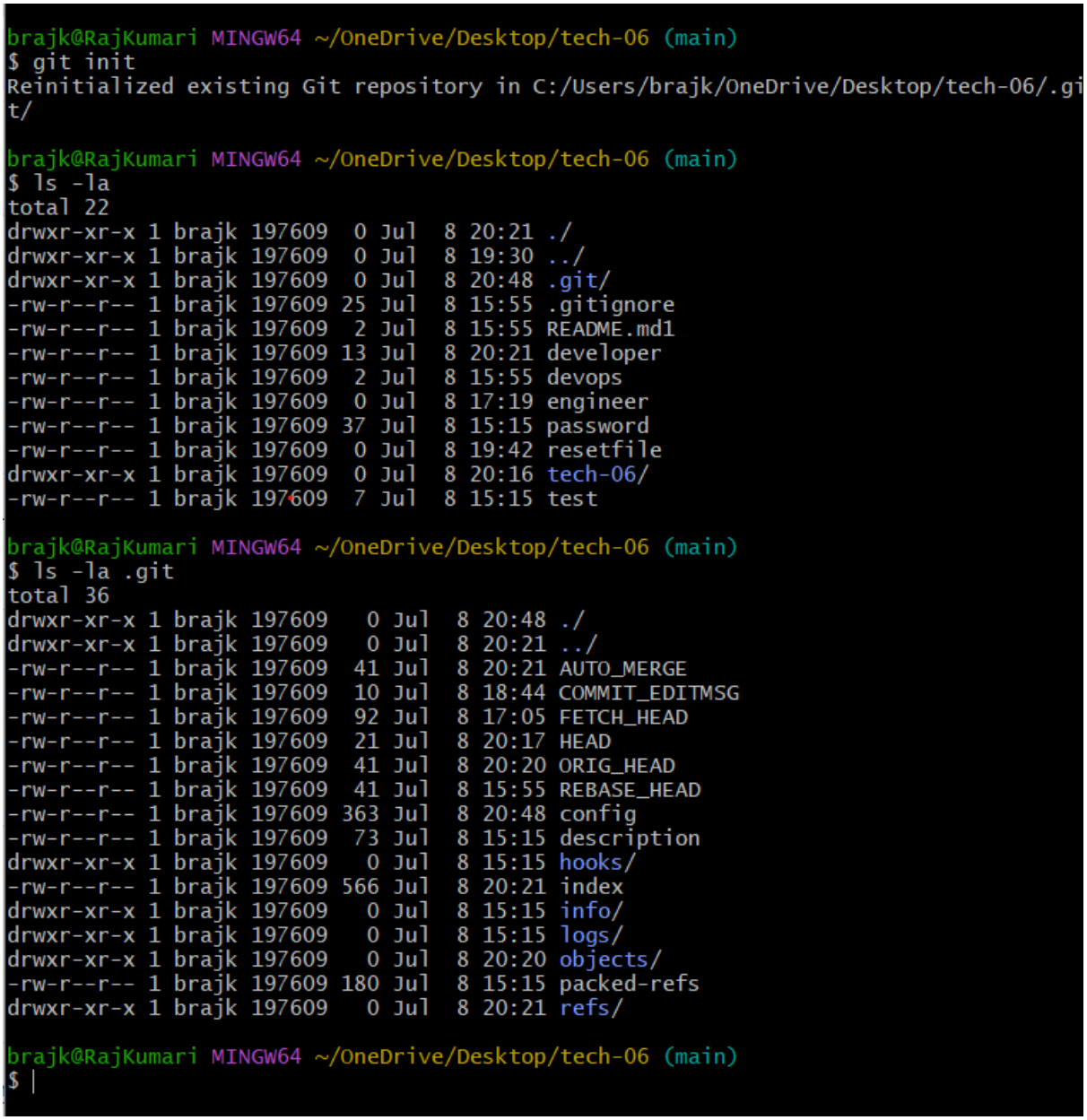
Purpose: The .git folder is the **hidden directory** that stores all the information Git needs to track changes, manage branches, store commits, and maintain the complete history of a repository. Without the .git folder, a directory is just a normal folder and is no longer a Git repository

What is the .git Folder?

- The .git folder is created automatically when you run:

$ git init

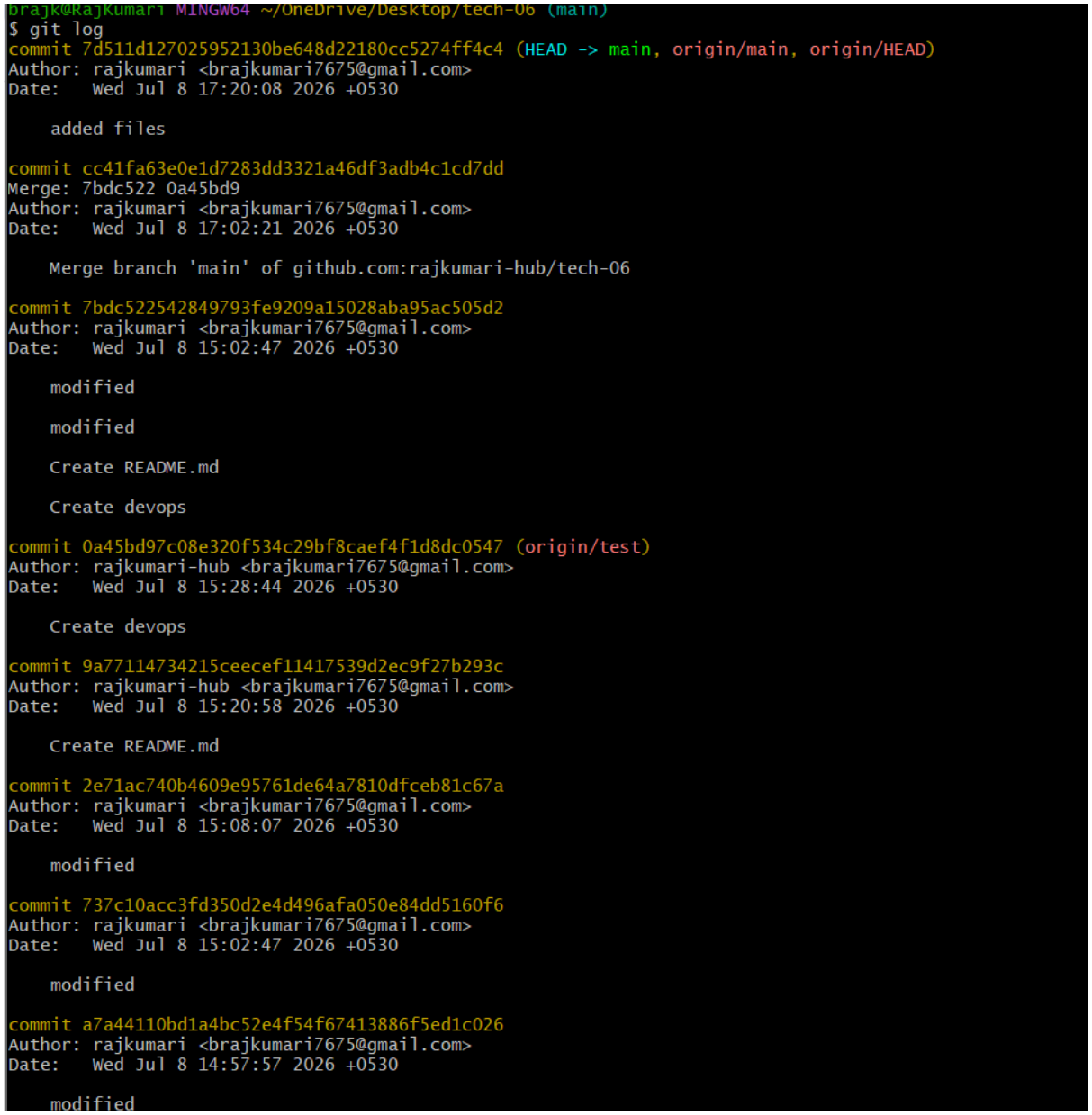
- It is located in the root directory of your Git repository.
- It is a hidden folder (because its name starts with a .)



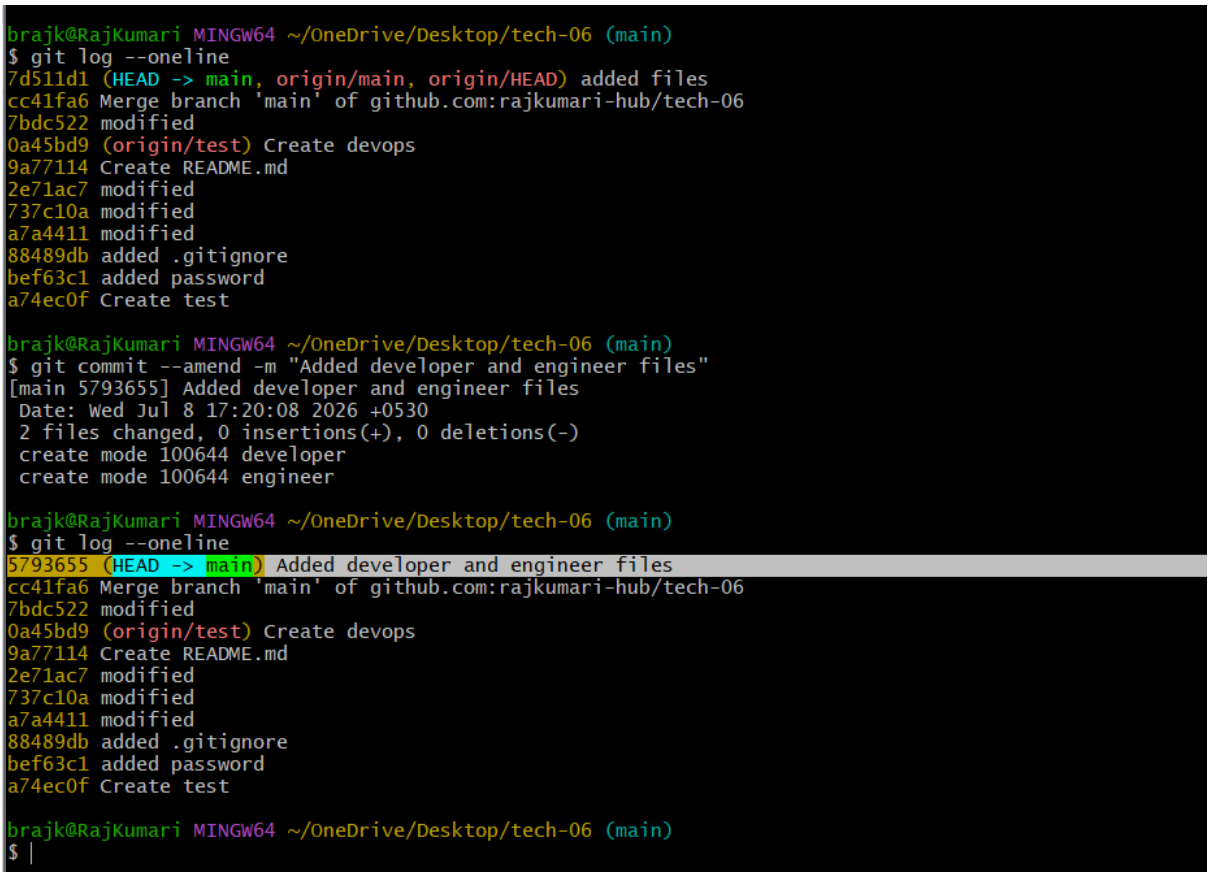
20. Check all the logs of git.

Objective: To view the commit history, branch movements, reference updates, and other Git activity using Git log commands.

Purpose: The purpose of Git logs is to track all changes made in a repository, including commits, branch operations, merges, authors, dates, and reference updates. Git logs help developers understand the project's history and troubleshoot issues.



21. Rename the commit message.



22. Merge multiple commits into single commit.

Objective: Combine multiple commits into one clean commit to keep the Git history simple and readable.

Purpose: This is useful when you have made several small commits while developing a feature and want to merge them into a single meaningful commit before pushing or creating a pull request.

$ git log -oneline

$ git rebase -i HEAD~4

```
pick 737c10a modified
squash 2e71ac7 modified
squash 9a77114 Create README.md
squash 0a45bd9 Create devops

# Rebase a7a4411..0a45bd9 onto a7a4411 (4 commands)
#
# Commands:
# p, pick <commit> = use commit
# r, reword <commit> = use commit, but edit the commit message
# e, edit <commit> = use commit, but stop for amending
# s, squash <commit> = use commit, but meld into previous commit
# f, fixup [-C | -c] <commit> = like "squash" but keep only the previous
#                    commit's log message, unless -C is used, in which case
#                    keep only this commit's message; -c is same as -C but
#                    opens the editor
# x, exec <command> = run command (the rest of the line) using shell
# b, break = stop here (continue rebase later with 'git rebase --continue')
# d, drop <commit> = remove commit
# l, label <label> = label current HEAD with a name
```

```
# This is a combination of 4 commits.
# This is the 1st commit message:

modified

# This is the commit message #2:

modified

# This is the commit message #3:

Create README.md
# This is the commit message #4:

Create devops

# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# Date:      Wed Jul 8 15:02:47 2026 +0530
#
# interactive rebase in progress; onto a7a4411
```

```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop/tech-06 (main)
$ git rebase -i HEAD~4
[detached HEAD 7bdc522] modified
 Date: Wed Jul 8 15:02:47 2026 +0530
 3 files changed, 4 insertions(+)
 create mode 100644 README.md1
 create mode 100644 devops
Successfully rebased and updated refs/heads/main.
```

```
commit 7bdc522542849793fe9209a15028aba95ac505d2 (HEAD -> main)
Author: rajkumari <brajkumari7675@gmail.com>
Date:   Wed Jul 8 15:02:47 2026 +0530

    modified

    modified

    Create README.md

    Create devops
```